

LTSpice Notch Quality Report

CROSS COUPLED_combined

Auto-detected swept parameters and quality-factor ($Q = f_0 / BW$ @ passband-3 dB) analysis

Run summary

Signal columns analyzed: 1

Total simulation steps : 25

V(act+,act-)

swept parameters: Cp, Itail

Report contents (per signal column):

- 1. All curves — Bode plot (magnitude + phase)
- 2. Magnitude grouped by each swept parameter (one subplot per fixed value)
- 3. Phase grouped by each swept parameter (one subplot per fixed value)
- 4. Quality-factor table for that column

End of report (extra analyses):

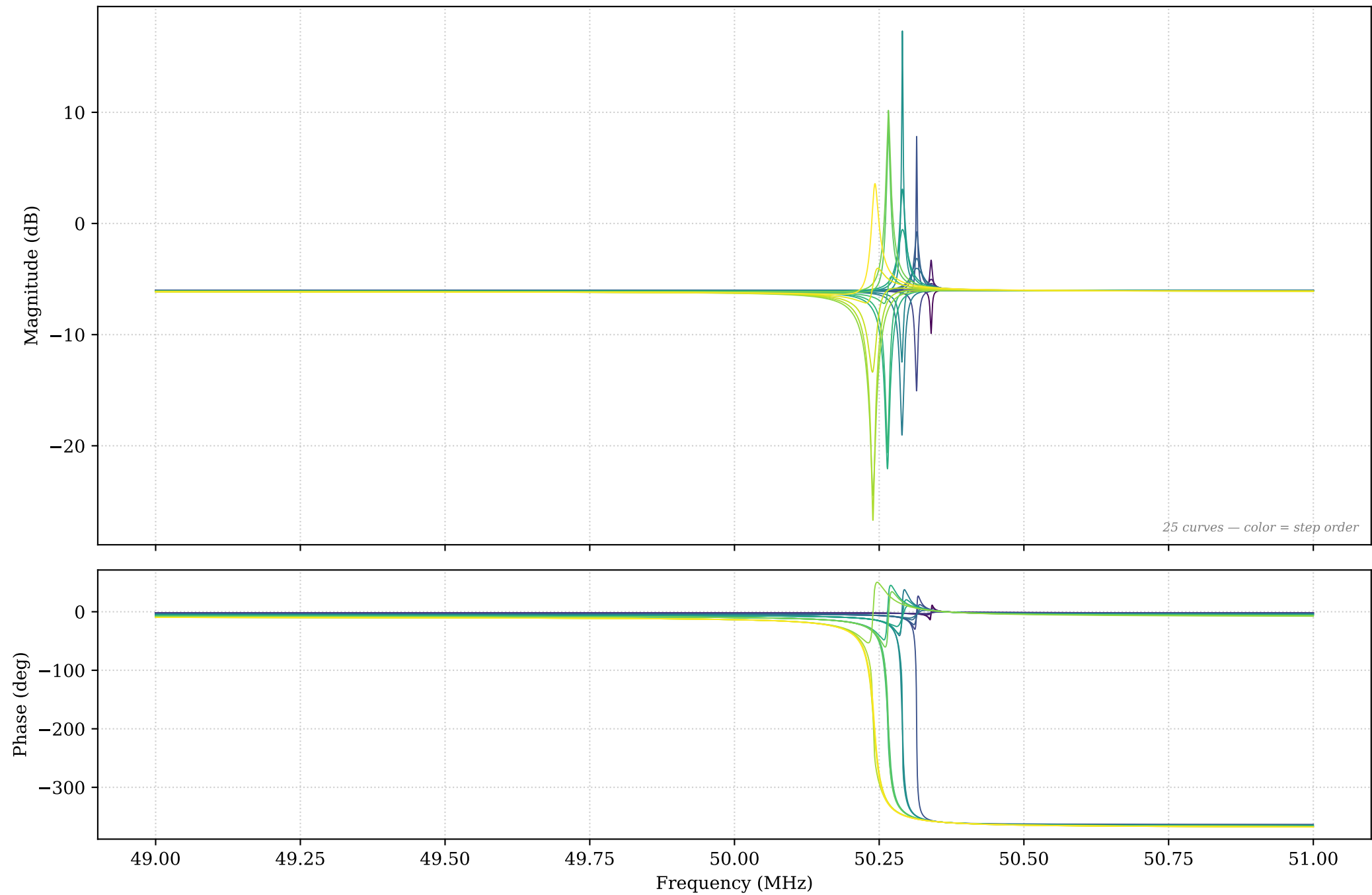
- A. Cross-signal comparison (act vs pas, if both present)
- B. Q vs each swept parameter (one line per other-param combo)
- C. BW vs each swept parameter
- D. Q heatmap for V(act): rows $\times$ columns swept params, colored
- E. Combined quality-factor table across all signals

Signal: $V(\text{act+}, \text{act-})$

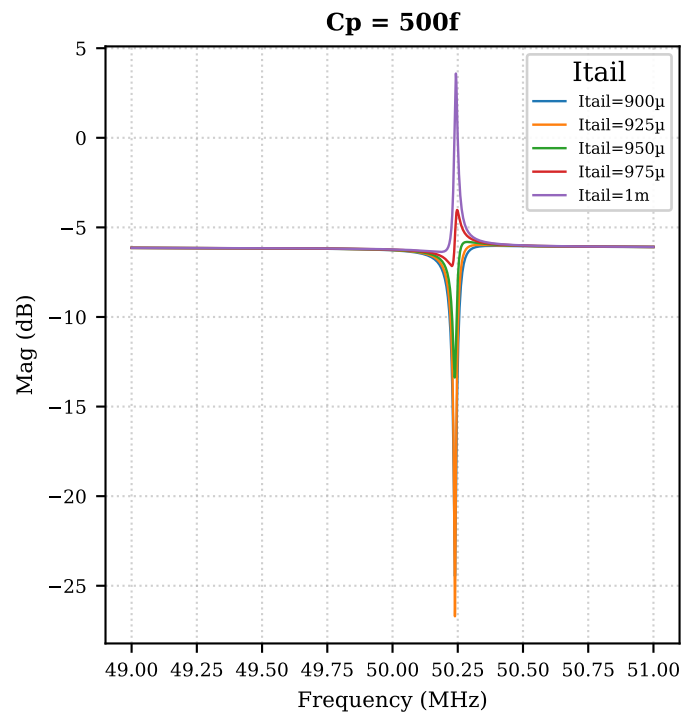
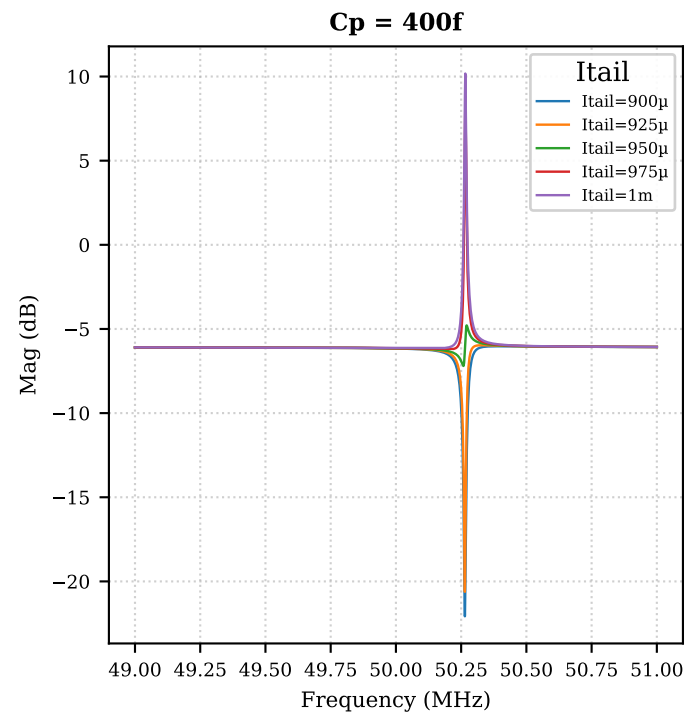
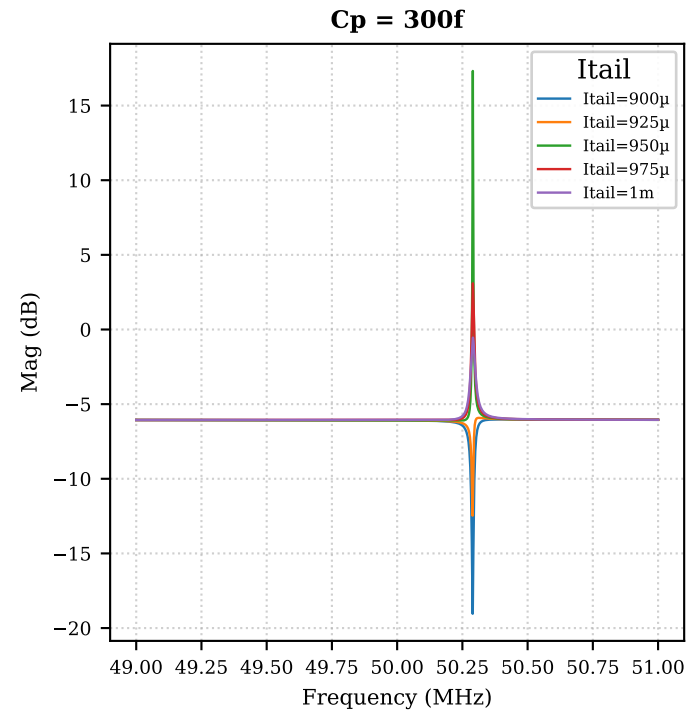
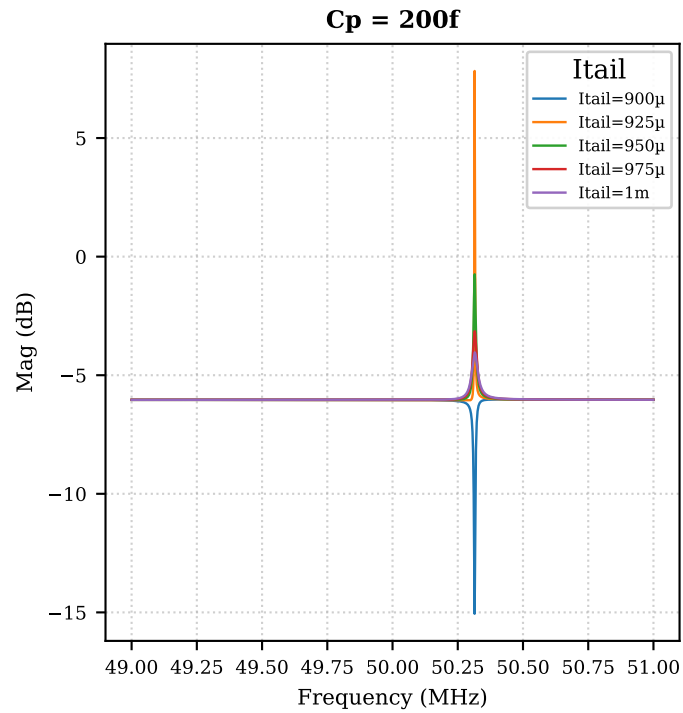
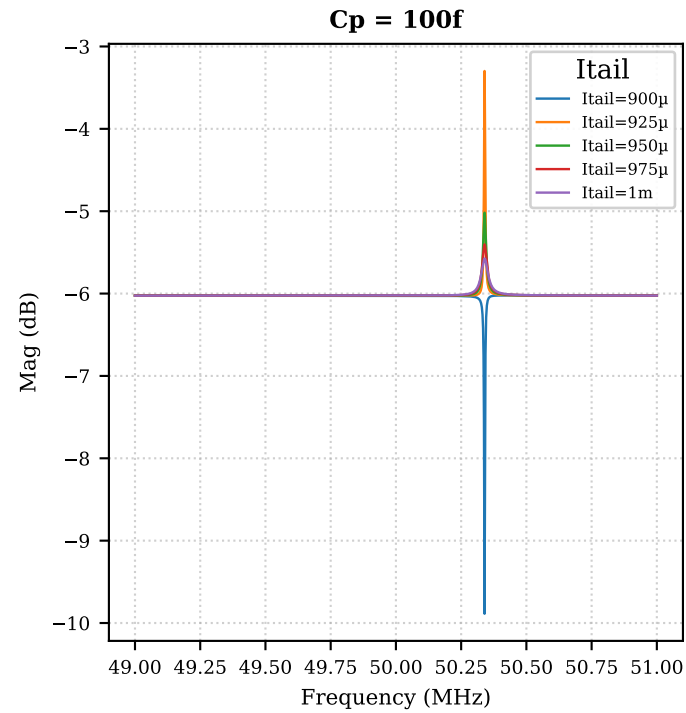
Swept parameters: C_p , I_{tail}

Steps: 25

V(act+,act-) — All curves (25 steps) — Bode

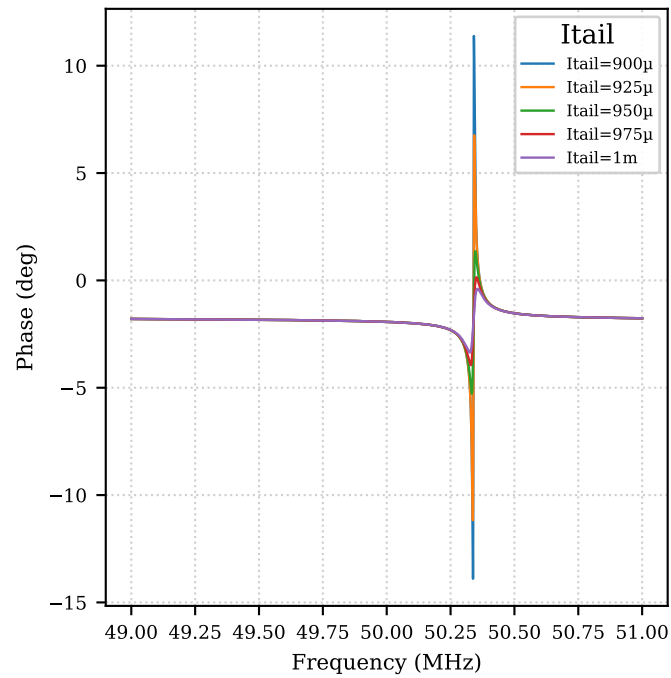


V(act+,act-) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)

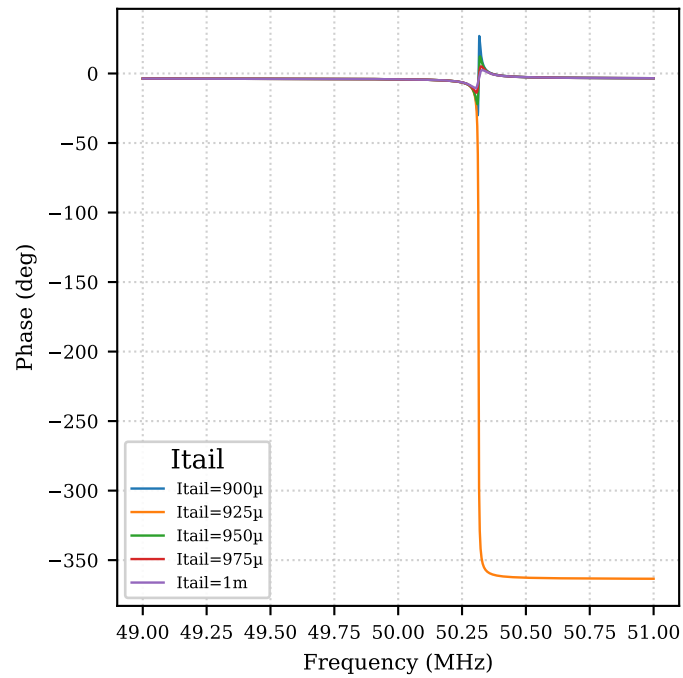


V(act+,act-) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)

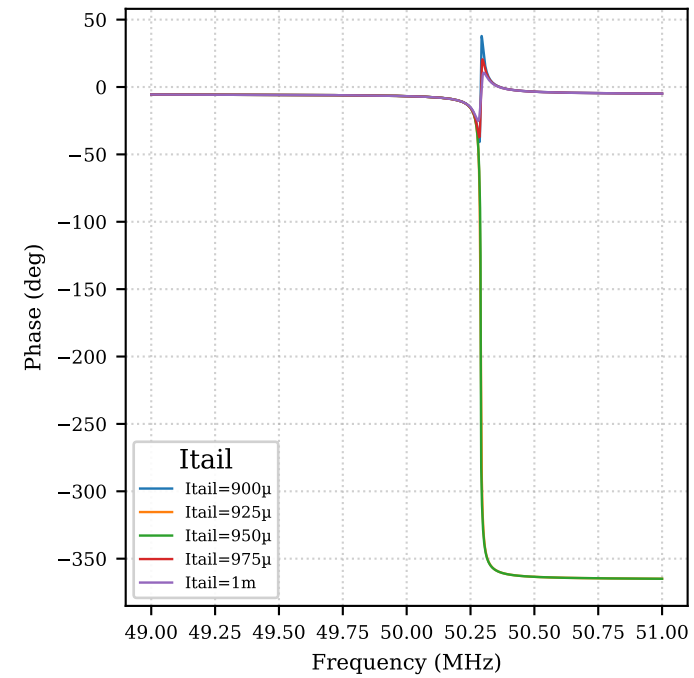
Cp = 100f



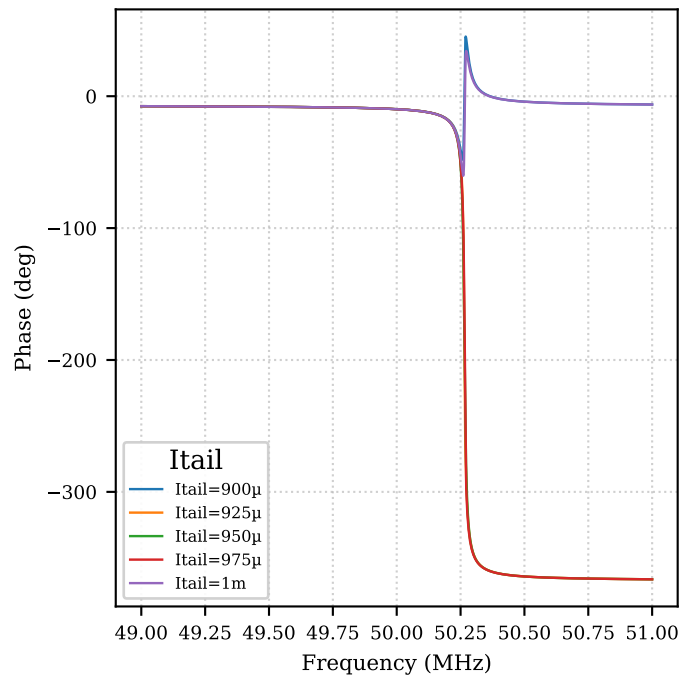
Cp = 200f



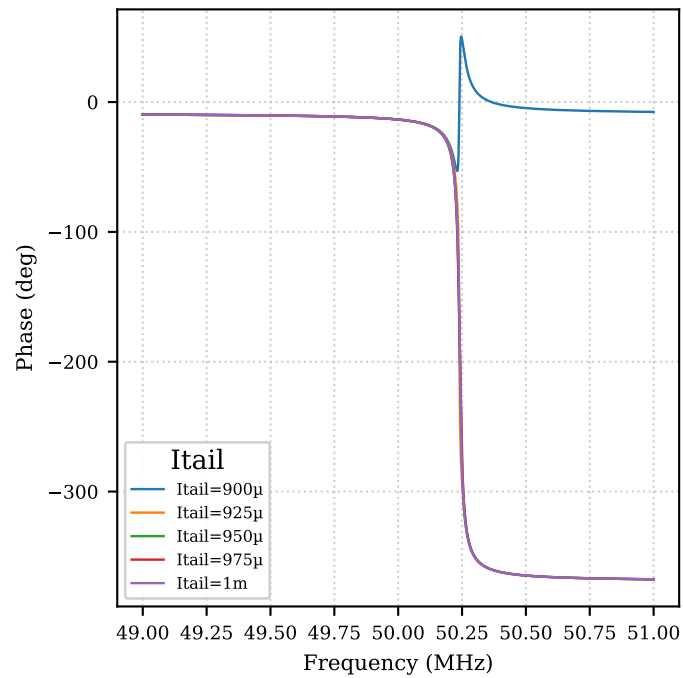
Cp = 300f



Cp = 400f

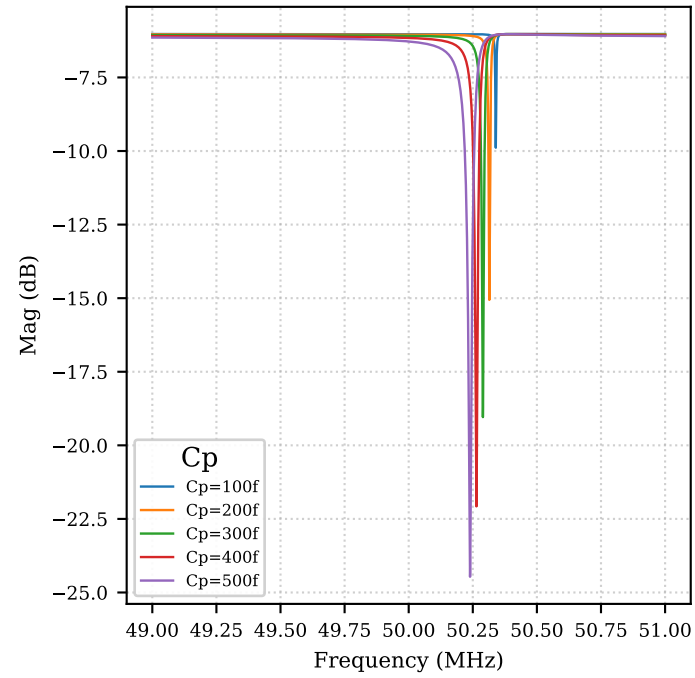


Cp = 500f

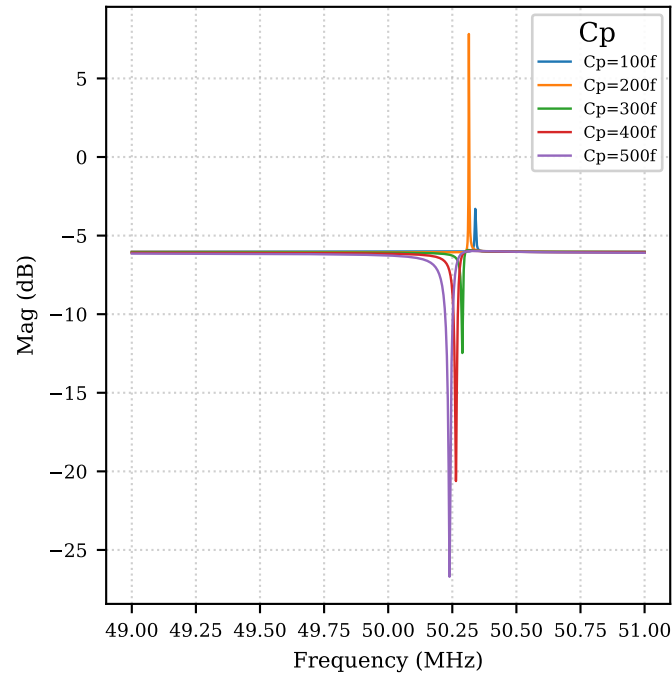


V(act+,act-) — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)

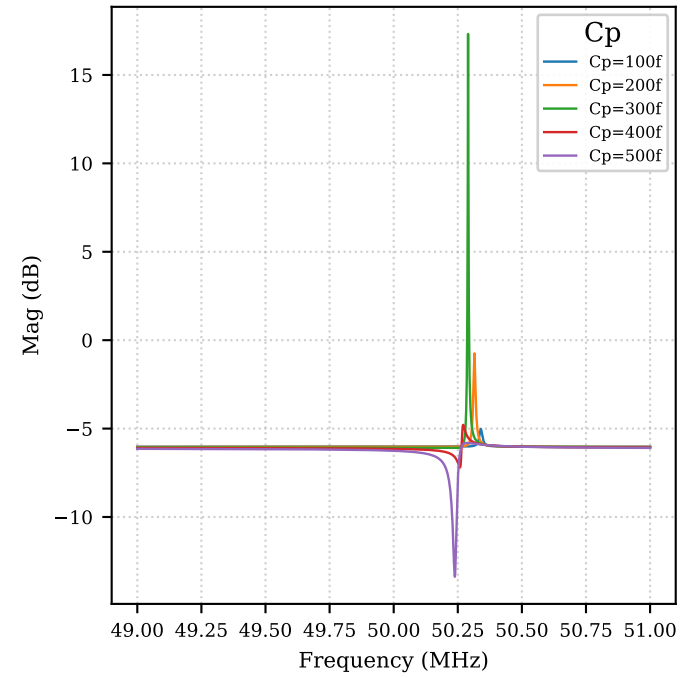
Itail = 900 μ



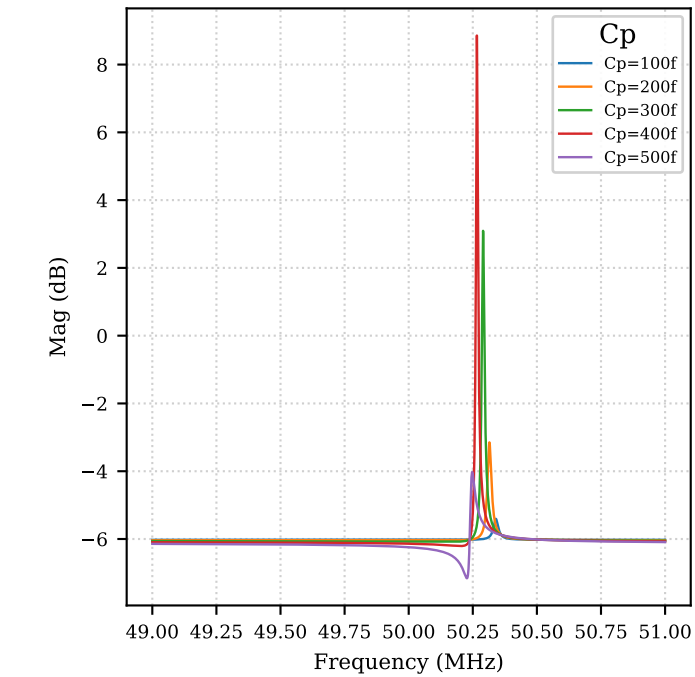
Itail = 925 μ



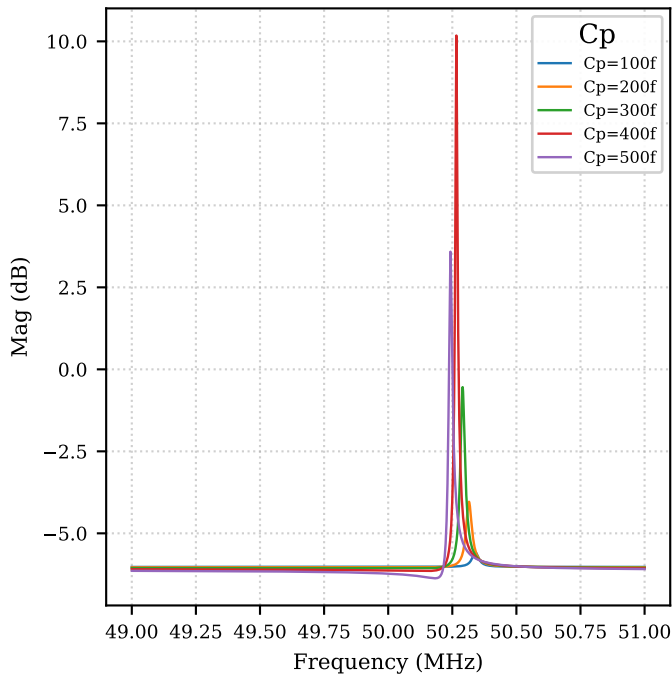
Itail = 950 μ



Itail = 975 μ

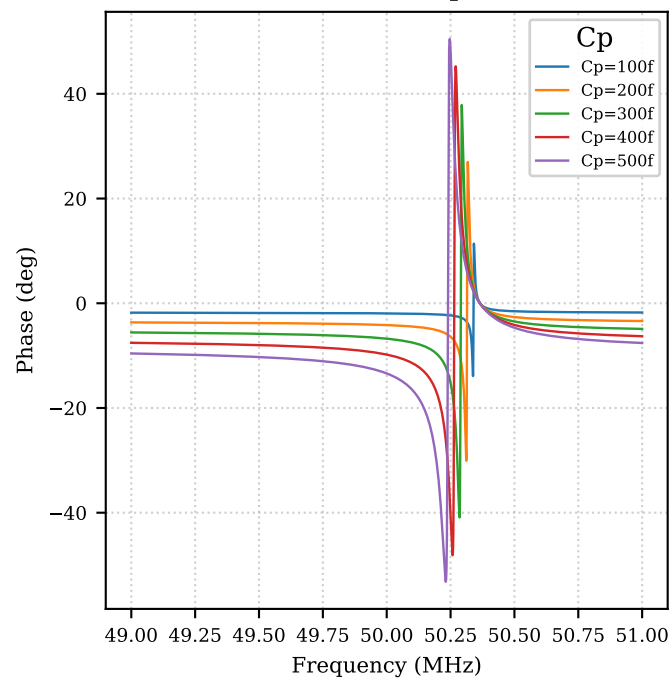


Itail = 1m

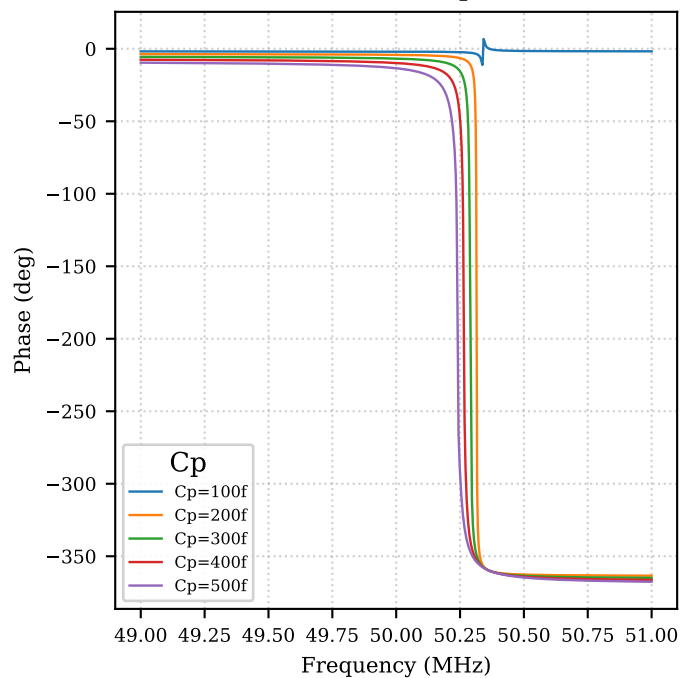


V(act+,act-) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)

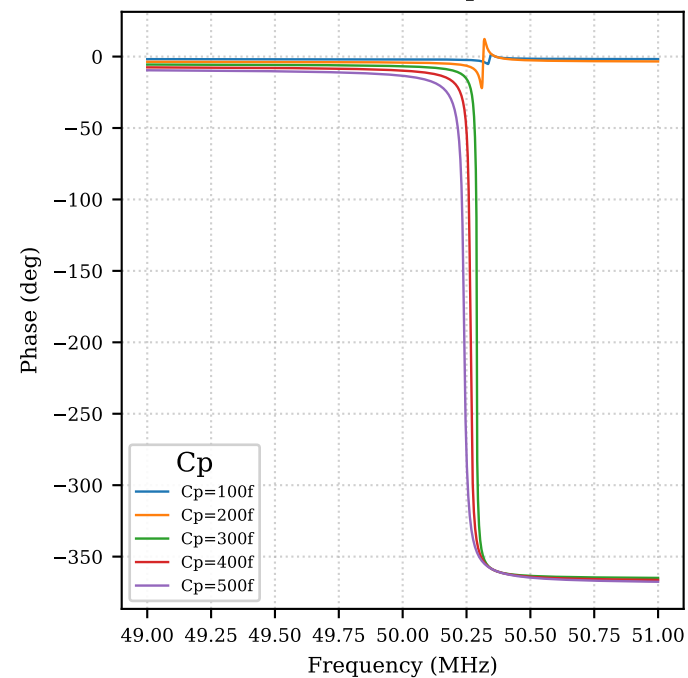
Itail = 900 μ



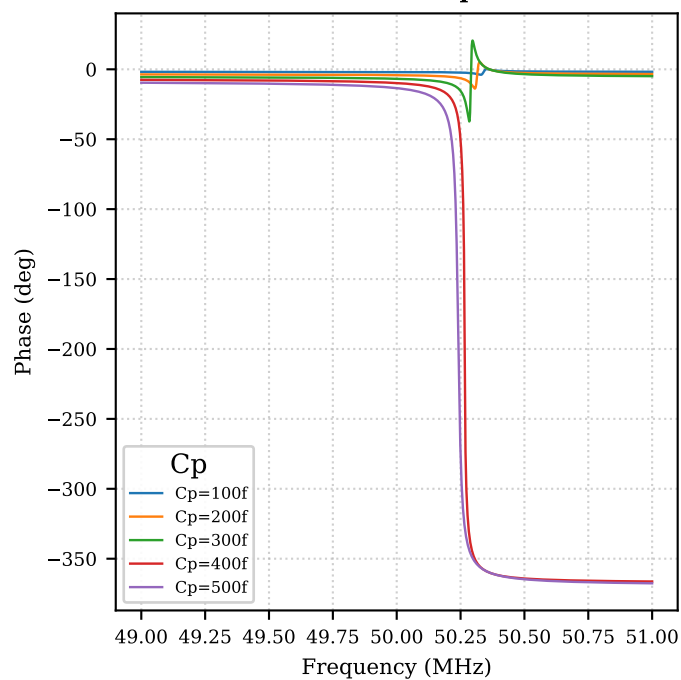
Itail = 925 μ



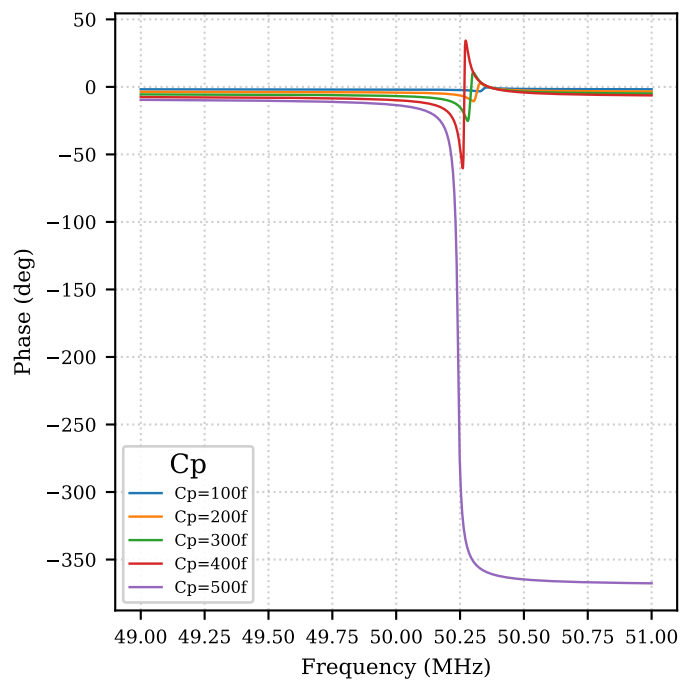
Itail = 950 μ



Itail = 975 μ



Itail = 1m

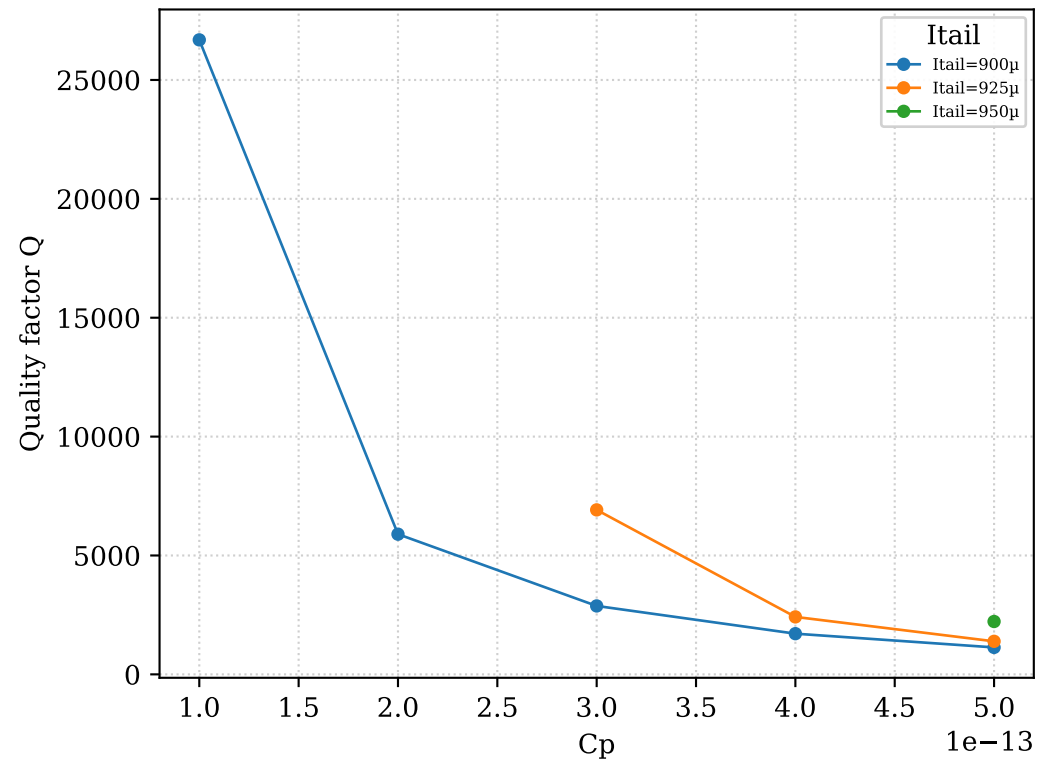


V(act+,act-) — Quality factor table

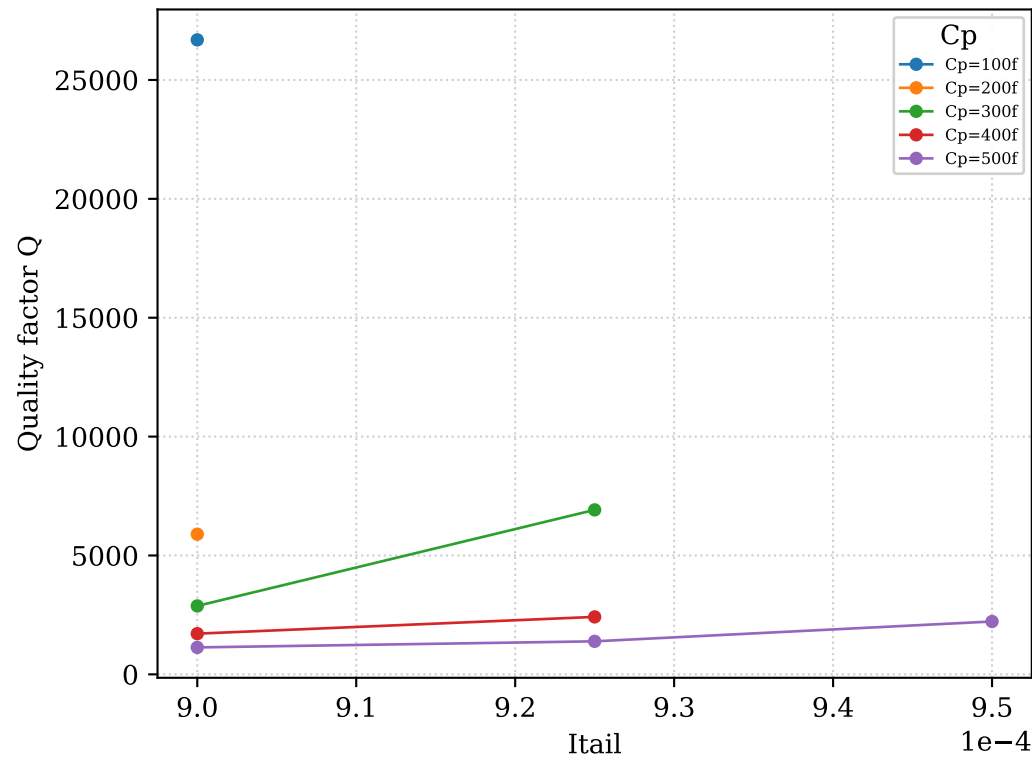
Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	900μ	50.34	-6.025	3.863	1886	2.669e+04	
V(act+,act-)	100f	925μ	50.23	-6.025	0.001513		notch shallower than 3 dB	
V(act+,act-)	100f	950μ	50.05	-6.025	0.0006926		notch shallower than 3 dB	
V(act+,act-)	100f	975μ	49.89	-6.025	0.0005179		notch shallower than 3 dB	
V(act+,act-)	100f	1m	49.76	-6.025	0.0004508		notch shallower than 3 dB	
V(act+,act-)	200f	900μ	50.31	-6.037	9.021	8534	5896	
V(act+,act-)	200f	925μ	50.29	-6.037	0.03855		notch shallower than 3 dB	
V(act+,act-)	200f	950μ	50.17	-6.037	0.008472		notch shallower than 3 dB	
V(act+,act-)	200f	975μ	50.07	-6.036	0.005298		notch shallower than 3 dB	
V(act+,act-)	200f	1m	49.97	-6.036	0.004136		notch shallower than 3 dB	
V(act+,act-)	300f	900μ	50.29	-6.057	12.98	1.747e+04	2878	
V(act+,act-)	300f	925μ	50.29	-6.057	6.414	7269	6918	
V(act+,act-)	300f	950μ	50.23	-6.056	0.05995		notch shallower than 3 dB	
V(act+,act-)	300f	975μ	50.15	-6.056	0.02716		notch shallower than 3 dB	
V(act+,act-)	300f	1m	50.08	-6.056	0.01862		notch shallower than 3 dB	
V(act+,act-)	400f	900μ	50.26	-6.085	15.99	2.937e+04	1711	
V(act+,act-)	400f	925μ	50.26	-6.084	14.53	2.08e+04	2417	
V(act+,act-)	400f	950μ	50.26	-6.083	1.118		notch shallower than 3 dB	
V(act+,act-)	400f	975μ	50.2	-6.083	0.1251		notch shallower than 3 dB	
V(act+,act-)	400f	1m	50.14	-6.082	0.06716		notch shallower than 3 dB	
V(act+,act-)	500f	900μ	50.24	-6.12	18.35	4.432e+04	1133	
V(act+,act-)	500f	925μ	50.24	-6.119	20.59	3.616e+04	1389	
V(act+,act-)	500f	950μ	50.24	-6.118	7.262	2.26e+04	2223	
V(act+,act-)	500f	975μ	50.23	-6.117	1.043		notch shallower than 3 dB	
V(act+,act-)	500f	1m	50.18	-6.116	0.255		notch shallower than 3 dB	

B. V(act+,act-) — Quality factor vs swept parameters

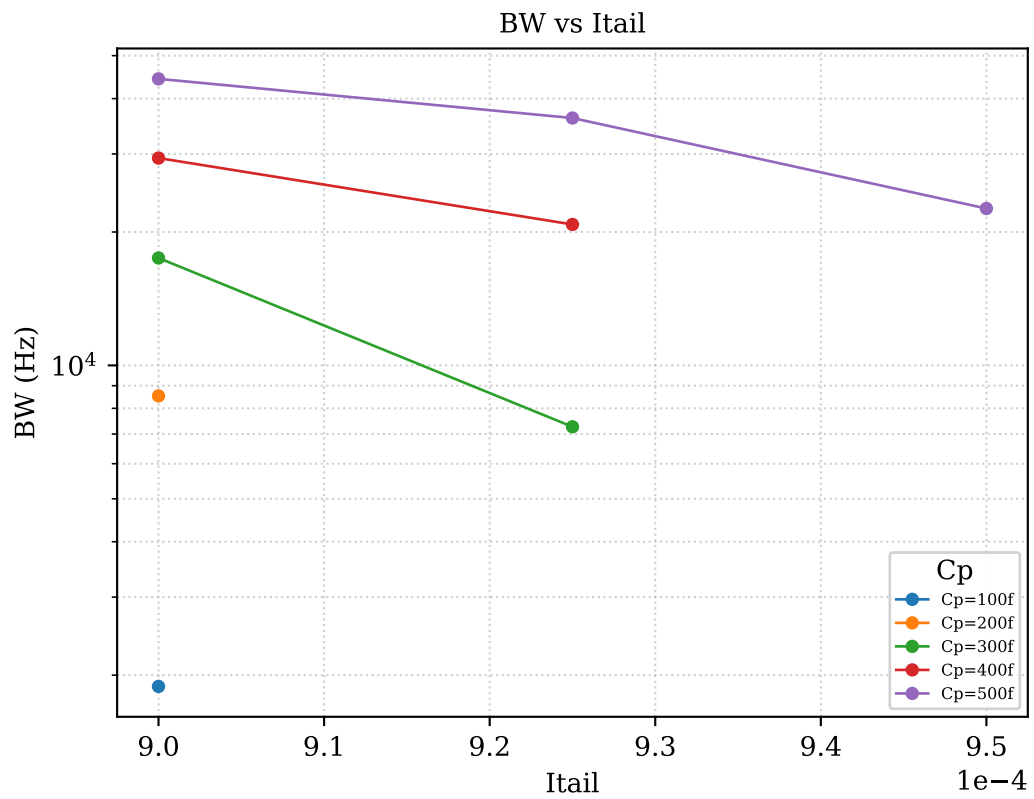
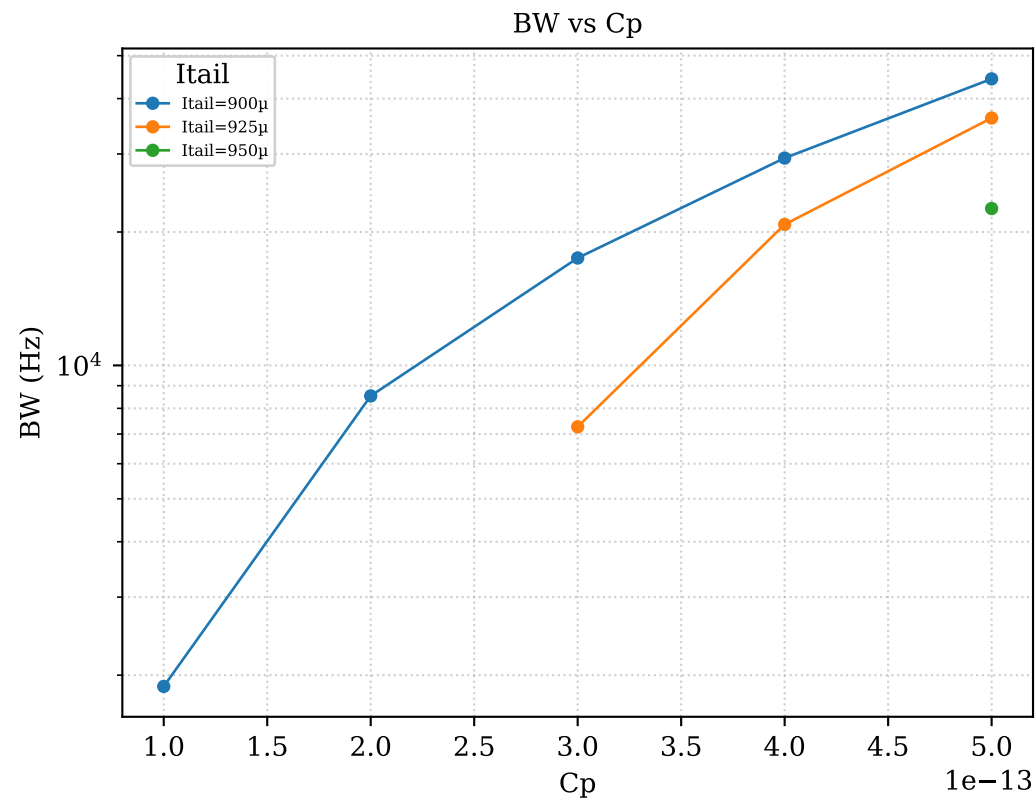
Q vs Cp



Q vs Itail



C. V(act+,act-) — Bandwidth (-3 dB) vs swept parameters



D. V(act+,act-) — Q heatmap (rows = Cp, columns = Itail)

	Itail=900μ	Itail=925μ	Itail=950μ	Itail=975μ	Itail=1m
Cp=100f	26687	no BW	no BW	no BW	no BW
Cp=200f	5896	no BW	no BW	no BW	no BW
Cp=300f	2878	6918	no BW	no BW	no BW
Cp=400f	1711	2417	no BW	no BW	no BW
Cp=500f	1133	1389	2223	no BW	no BW

Red = BW could not be measured | Yellow = Q computed ($Q \leq 1400$) | Green = $Q > 1400$

E. Combined quality-factor table — all signals

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	900μ	50.34	-6.025	3.863	1886	2.669e+04	
V(act+,act-)	100f	925μ	50.23	-6.025	0.001513		notch shallower than 3 d	
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